Agent Drift and Guardrail Dilution: Natural Language versus Deterministic Templates

Notes on the tokenizer-agnostic dilution model

Abstract

A next-token predictor cannot distinguish a natural-language instruction from a function call: the tokenizer assigns each exactly one token and one unit of attention. Yet only function calls *execute*. A buffer of N natural-language (NL) tokens therefore dilutes the original user query / guardrail by the attention weight it receives; a buffer of function calls, or a deterministic table lookup, does not. We give a one-line model for this dilution, show that it is exponential (or sub-exponential) in buffer length and in matryoshka nesting depth, and compare a heavy NL chat template with a minimal Jinja template.

1. Setup

Let L be the context length, with

$$\text{position } 0 \;=\; \text{original user query / guardrail}, \qquad \text{positions } 1,\dots,L-1 \;=\; \text{agent buffer}.$$

Write n_{FN} for the number of function-call tokens in the buffer and $n_{\text{NL}} = L - 1 - n_{\text{FN}}$ for the number of natural-language tokens. An attention kernel yields weights $w_0, \dots, w_{L-1}$ summing to 1.

The guardrail signal at position 0 is

$$h_{\text{safe}} \;=\; w_0 \;+\; \rho\, n_{\text{FN}}, \tag{1}$$

where $\rho \in (0, 1]$ is the re-injection strength of a function call (a call *executes*; an NL token merely *attends*). The natural-language noise is

$$\sigma^2_{\text{noise}} \;=\; \sigma^2_{\text{NL}} \sum_{i \in \text{NL}} w_i^2. \tag{2}$$

The guardrail signal-to-noise ratio and the implied safe-action probability are

$$\text{SNR}(L, n_{\text{FN}}) \;=\; \frac{h_{\text{safe}}}{\sigma_{\text{noise}}}, \qquad P_{\text{safe}} \;=\; \Phi(\text{SNR})\,, \qquad P_{\text{mis}} \;=\; 1 - P_{\text{safe}}, \tag{3}$$

where Φ is the standard normal CDF.

2. Three regimes

For a pure natural-language buffer ($n_{\text{FN}} = 0$), the guardrail weight w_0 decays according to the kernel:

kernel	$w_0(L)$	SNR(L) asymptotics
uniform	$1/L$	$\Theta(\sqrt{L})$
power-law	$\propto L^{-\beta}$	$\Theta(L^{\beta-1/2})$
exponential	$\propto e^{-\lambda L}$	$\Theta(e^{\lambda L})$

In every case SNR $\to 0$ as $L \to \infty$, hence $P_{\text{safe}} \to \frac{1}{2}$: the model's behaviour becomes statistically indistinguishable from random *relative to the guardrail*, even though no weight has changed and no instruction has been violated. The drift is *dilution*, not misalignment of intent.

3. Matryoshka nesting

A real multi-agent system nests instructions: user query → `AGENTS.md` → sub-agent prompt → $\cdots$. With k nesting levels and M NL tokens per level,

$$L_k \;=\; 1 + kM, \qquad \mathrm{SNR}(k) \;\simeq\; \mathrm{SNR}(0)\, e^{-\lambda k M}. \tag{4}$$

The decay is thus *exponential in the nesting depth*. Because each sub-agent may have its own context window, the original query can fall outside the window entirely, in which case SNR $= 0$ and $P_{\mathrm{safe}} = \frac{1}{2}$ exactly.

4. Function calls restore the signal

Substituting n_{FN} into (1)–(2) yields

$$\mathrm{SNR}(L, n_{\mathrm{FN}}) \;=\; \frac{w_0 + \rho\, n_{\mathrm{FN}}}{\sigma_{\mathrm{NL}} \sqrt{\sum_{\mathrm{NL}} w_i^2}}. \tag{5}$$

Because h_{safe} grows linearly in n_{FN} while the noise grows only as the L^2-norm of the remaining NL weights, a small number of function calls placed *near the action* keeps SNR bounded away from zero as $L \to \infty$. A deterministic table lookup (*e.g.* the MOSB read-only buffer used in the wider pipeline) is even stronger: it is a $O(1)$ index lookup with SNR $= 1$ by construction, because no attention competition occurs.

5. Chat-template comparison

Table 1 compares a heavy NL template (prose instructions, examples, boilerplate) with a minimal Jinja template (short system prompt, three function calls, no prose). Numbers are from the accompanying simulation with $\lambda = 0.05$, $\sigma_{\mathrm{NL}} = 0.5$, $\rho = 0.8$.

template	L	n_{FN}	SNR	P_{safe}
heavy NL prose	150	0	1.6×10^{-2}	0.506
minimal Jinja	10	3	7.8×10^{0}	1.000
improvement	$\frac{1}{15}$	—	$\sim 5 \times 10^{2}$	—

Table 1: Heavy NL template versus minimal Jinja template. The minimal template is $\sim$500$\times$ better in SNR at 1/15 of the context length.

The difference is entirely structural. The tokenizer sees the two prompts as equivalent streams of unit-cost tokens; the model sees one as a sequence of competing observations and the other as a sequence of executing function calls. Replacing prose guardrails with function calls or table lookups placed near the action is therefore the correct fix — not a longer system prompt, and not a re-weighted tokenizer, both of which leave the dilution exponent unchanged.

6. Summary

- Natural-language tokens and function-call tokens cost the same to the tokenizer but behave differently in the network.
- A pure NL buffer dilutes the guardrail at rate $w_0(L) \propto e^{-\lambda L}$ (exponential kernel), $L^{-\beta}$ (power-law), or $1/L$ (uniform).

- Nesting multiplies the effect: $\mathrm{SNR}(k) \simeq \mathrm{SNR}(0)\, e^{-\lambda k M}$.
- Function calls re-inject at full strength and keep SNR bounded; deterministic tables have $\mathrm{SNR} = 1$.
- The observed mis-alignment is $P_{\mathrm{safe}} \to \frac{1}{2}$ — statistical drift, not adversarial intent.